

## Expose IX. Neron models and monodromy, continued

### 6.2. On the fixed part outside the semi-stable case

Let us return to the largest  $p$ -divisible subgroup  $A_o^{\circ p\text{-div}}$  of the neutral component of the special fibre, and write

$$F_o = p^\infty(A_o^\circ).$$

It is the largest Barsotti–Tate  $p$ -subgroup of  $A_o$ , equivalently of  $A_o^\circ$ . It is natural to ask whether this group can be lifted to a Barsotti–Tate subgroup of the Neron model  $A$ , playing the role which the fixed part  $T_p(A)^f$  plays in the prime-to- $p$  theory and in the semi-stable case.

When such a lift exists, it is unique. Moreover the resulting sub-pro-groups in  $T_p(A)$  and  $T_p(A')$ , which deserve again to be called fixed parts, satisfy the same orthogonality theorem as in the semi-stable case. The fixed part can still be characterized by the condition already used above for elements of the Tate module.

However this question does not have an affirmative answer in general. By the Serre–Tate lifting theorem, when  $S$  is complete the existence of the fixed part is equivalent to the existence of compatible infinitesimal liftings of the relevant special-fibre Barsotti–Tate group. Equivalently, one would need a smooth group scheme over  $S$ , an extension of an abelian scheme by a torus, whose special fibre is the prescribed group and which maps formally to the Neron model extending the inclusion on the special fibre. In particular, when the toric part is zero, this is the same as asking for an abelian scheme lifting the special fibre, together with a homomorphism into the Neron model. Raynaud’s examples show that this is not always possible.

Here is the form of the example used in the source. Let  $S' \rightarrow S$  be the normalization in a finite separable ramified extension  $K'/K$ , and let  $A'$  be an abelian scheme over  $S'$ . The Weil restriction

$$A = \text{Res}_{S'/S}(A')$$

is Neronian over  $S$ : it identifies with the Neron model of its generic fibre. The neutral special fibre is an extension of an abelian scheme by a connected unipotent group, non-zero as soon as the extension is ramified and  $A' \neq 0$ . Taking  $K'/K$  quadratic and  $A'$  an elliptic curve over  $S'$ , one arranges that  $A_K$  is simple over  $K$  by choosing the invariant  $j$  in such a way that it does not descend algebraically from  $K$ . This gives precisely a case where the desired fixed part cannot be constructed.

### 6.3. Essential fixed and toric parts

Assume now only that  $S$  is henselian. Choose a finite Galois extension  $K'/K$  over which  $A_K$  acquires semi-stable reduction, and let  $S'$  be the normalization of  $S$  in  $K'$ . The semi-stable theory over  $S'$  gives the fixed part and the toric part of the corresponding pro-Barsotti–Tate group. Descending by Galois invariants gives two strict sub-pro-groups of  $T_p(A)$ , denoted

$$T_p(A)^{ef} \subset T_p(A), \quad T_p(A)^{et} \subset T_p(A), \quad (6.3.1)$$

called respectively the *essentially fixed part* and the *essentially toric part*. These definitions are independent of the chosen semi-stable extension. The toric part is the orthogonal of the fixed part for the canonical pairing, and the same assertion remains true after descent.

If  $A_K$  has essential good reduction, the essentially fixed part is the whole Tate group. If  $A_K$  has no such fixed contribution, the essentially toric part controls the monodromy part of the representation. In residue characteristic  $p$ , the trace functions attached to these objects are defined by passing, after finite extension, to the Dieudonné module of the special fibre. For a Barsotti–Tate group  $Y$  over the residue field one associates a finite free module  $M$  over the Witt vectors  $W(k')$ . The Galois group acts semilinearly on  $M$ , and the inertia trace thereby obtained agrees with the trace function previously defined on  $T_\ell$  for  $\ell \neq p$ , once the usual comparison is made in the semi-stable situation. The construction used in the proof of 4.3.2 also gives the corresponding trace function on the higher inertia quotients.

## 7. Raynaud’s extension attached to an abelian variety with semi-stable reduction

### 7.1. The basic construction

Let  $S = \operatorname{Spec} V$ , where  $V$  is a noetherian ring complete and separated for the topology defined by an ideal  $\mathfrak{m}$ . Let  $A$  be a smooth group scheme over  $S$ , let  $A^\circ$  be its neutral component, and suppose that  $A_\circ/A_\circ^\circ$  is finite over the closed fibre and that  $A_\circ^\circ$  is an extension of an abelian scheme  $B_\circ$  by an isotrivial torus  $T_\circ$ . One associates to  $A$  a system of extensions which, in the semi-stable case, is Raynaud’s extension of the abelian variety.

The construction begins with a torus  $T$  over  $S$ , lifting the torus  $T_\circ$ , and with an abelian scheme  $B$  over  $S$ , lifting  $B_\circ$ , together with an extension

$$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0$$

whose special fibre maps to the neutral component of  $A_\circ$ . The source constructs this extension by applying the algebraization and extension results established in Exposé VIII. For the dual abelian scheme one obtains a second extension

$$0 \longrightarrow T' \longrightarrow A'^\natural \longrightarrow B' \longrightarrow 0,$$

and the two extensions are related by the Poincaré biextension.

The first statements establish that, under the stated hypotheses,  $A^\natural$  is algebraizable. If  $S$  is normal, the corresponding abelian scheme  $B$  is projective over  $S$ . Indeed, a relatively ample invertible sheaf on  $A$  gives a birigidified invertible sheaf on  $A \times_S A$ ; the absence of unipotent rank in the fibres allows the construction of a symmetric biextension, and the amplitude criterion of Raynaud implies projectivity. The normality hypothesis can then be removed by reduction to the normal case.

**Proposition 7.1.5.** With the preceding notation, if  $S$  is normal, the extension  $A^\natural$  is algebraizable, and the abelian scheme  $B$  from which it comes is projective over  $S$ . The same algebraizability conclusion remains true without normality after passage to the normal case.

### 7.2. The category of semi-stable smooth group schemes

Let  $\mathcal{C}$  be the category of smooth finite-type group schemes  $A$  over  $S$  such that  $A/A^\circ$  is finite étale over the closed fibre and  $A^\circ$  is an extension of an abelian scheme  $B$  by an isotrivial torus  $T$ . Raynaud’s analysis identifies this category with a category of data consisting of

$$(B, T, Y, \text{extension data and component data}),$$

where  $Y$  is the character group of the torus and the component group acts through finite étale data. The exact form is used to compare the group scheme  $A$  with its dual construction. The lemma in the source shows that this description is stable under the operations which will be used below: duality, passage to finite étale covers, and base change between complete traits.

The resulting extension  $A^\natural$  has generic fibre

$$A_K^\natural = (A^\natural)_K, \quad (7.6.1)$$

and is itself an extension of the finite étale component group by the neutral extension of the abelian variety by a torus. For every prime  $\ell$  one obtains canonical identifications

$$T_\ell(A_K^\natural) \simeq T_\ell(A_K^{\natural\circ}) \simeq T_\ell(A_K)^f, \quad T_\ell(T_K) \simeq T_\ell(A_K)^t. \quad (7.6.2)$$

The formation of the neutral Raynaud extension commutes with every base change of complete traits.

### 7.3–7.6. Duality and polarizations

The Poincaré biextension on  $(B, B')$  pulls back to a biextension on  $(A^\natural, A'^\natural)$ . It yields a duality between the toric and cotoric pieces, and therefore a pairing between the lattices of characters and cocharacters. If a divisor correspondence  $W_K$  on  $(A_K, A'_K)$  is a perfect duality, the same is true for the corresponding correspondence  $W^\natural$  on Raynaud extensions. If  $W_K$  is attached to a polarization of  $A_K$ , then  $W^\natural$  is attached to a polarization of  $A^\natural$ .

The formation of  $W^\natural$  is compatible with inverse images of correspondences under homomorphisms of abelian schemes

$$\overline{A}_K \rightarrow A_K, \quad \overline{A}'_K \rightarrow A'_K.$$

This follows from the intrinsic characterization of the biextension and the compatibility of the Poincaré biextension with pullback. The assertion about polarizations is the one used in 7.1.5 to prove algebraizability and projectivity.

## 8. Passage to the limit over finite extensions

Let  $K'/K$  run through finite separable extensions inside a fixed separable closure, and write  $S'$  for the corresponding normalization of the trait. Applying Raynaud's construction after base change gives group schemes  $A_{K,K'}^\natural$  over  $K$ . If  $K''/K'$  is another such extension, there is a canonical open immersion

$$A_{K,K'}^\natural \hookrightarrow A_{K,K''}^\natural. \quad (8.3.1)$$

The quotient

$$\Phi_{K,K'} = A_{K,K'}^\natural / A_{K,K'}^{\natural\circ} \quad (8.3.2)$$

is finite étale over  $K$ , and the transition morphisms are compatible with the open immersions. Thus one may set

$$A_K^b = \varinjlim_{K'} A_{K,K'}^\natural \quad (8.3.3)$$

in the category of fppf sheaves on  $K$ . This limit is in fact represented by a group scheme over  $K$ , generally not quasi-compact. With the same notation one has

$$A_K^{b\circ} = A_K^{\natural\circ}, \quad A_K^b / A_K^{b\circ} \simeq \varinjlim_{K'} \Phi_{K,K'}. \quad (8.3.4)$$

The latter group is identified later with the constant sheaf  $M \otimes \mathbb{Q}/\mathbb{Z}$ , where  $M$  is the monodromy lattice. This prepares the definition of the monodromy pairing in the following section.

# SGA 7-I

## Exposé IX. Néron Models and Monodromy Monodromy Pairings and Component Groups

### 9. Definition of the monodromy pairing

#### 9.1. The setting

In this section the trait  $S$  is henselian and the abelian variety  $A_K$  has semi-stable reduction over  $S$ . Equivalently, the same is true for the dual abelian variety  $A'_K$ . We shall use the twisted constant  $\mathbb{Z}$ -modules  $M$  and  $M'$  attached to the maximal tori of the special fibres of the connected Néron models of  $A_K$  and  $A'_K$ . After a suitable finite étale extension of  $S$  these become constant free groups, non-canonically isomorphic to  $\mathbb{Z}^r$ , where  $r$  is the common rank of the toric parts.

For each prime  $\ell$ , put

$$M_\ell = M \otimes \mathbb{Z}_\ell, \quad M'_\ell = M' \otimes \mathbb{Z}_\ell. \quad (9.1.1)$$

The geometric meaning of these sheaves is given by the formulae of the preceding sections, where they occur as the Tate modules of the toric parts. Our aim is to define a canonical bilinear form, the *monodromy pairing*,

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell. \quad (9.1.2)$$

It will be defined by means of the Barsotti-Tate pro- $\ell$ -groups  $T_\ell(A^0)$ ,  $T_\ell(A'^0)$ , and even by means of the generic fibre  $T_\ell(A_K)$  of the first of these groups. When  $\ell = p$ , this last description uses Tate's theorem for homomorphisms of Barsotti-Tate groups; for the present purposes it is available in the equalities of characteristic to which it is applied below.

When  $\ell \neq p$ , the same construction may be expressed in purely Galois-theoretic terms. If

$$U = T_\ell(A_K(\bar{K})) \quad (9.1.3)$$

with its action of the inertia group  $I$ , then  $u_\ell$  is the datum which measures the unipotent monodromy of  $I$  on  $U$ .

#### 9.2. The case $\ell \neq p$

Assume first that  $\ell \neq p$ . We use the notation introduced earlier for the exact sequence of  $I$ -modules attached to the semi-stable reduction of  $A_K$ . Up to torsion, the relation which defines the toric submodule says that it is generated by elements of the form

$$gx - x \quad (x \in U, g \in I). \quad (9.2.1)$$

Thus for each  $g \in I$  one has

$$gx = x + u(g)(x), \quad (9.2.2)$$

where  $u(g)$  is a homomorphism from  $U$  to the toric part, vanishing on that toric part. Equivalently,

$$u(g) \in \text{Hom}(U/V, V), \quad (9.2.3)$$

where  $V$  denotes the submodule on which the transvections are parallel. In the semi-stable case the operation of every  $g$  is a transvection parallel to  $V$ , and composition of such transvections corresponds to addition of the associated homomorphisms. Hence the action of inertia is described by a homomorphism

$$I \longrightarrow \text{Hom}(U/V, V). \quad (9.2.4)$$

By the orthogonality theorem applied both to  $A_K$  and to  $A'_K$ , one identifies  $U/V$  with the Tate twist of the toric Tate module of  $A'_K$ . Consequently the homomorphism (9.2.4) may be interpreted as a canonical homomorphism

$$M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell, \quad (9.2.5)$$

or as an element

$$u_\ell \in \text{Hom}(M_\ell \otimes M'_\ell, \mathbb{Z}_\ell). \quad (9.2.6)$$

This is the element announced in (9.1.2) in the case  $\ell \neq p$ . It is invariant under transport of structure, hence is defined over the original residue field and not merely after passing to a separably closed residue field.

To define (9.1.2) without excluding  $\ell = p$ , we next recall a purely homological construction.

### 9.3. Mixed extensions

Let  $\mathcal{C}$  be an abelian category and let

$$P, \quad R, \quad Q \quad (9.3.1)$$

be three objects of  $\mathcal{C}$ . We are interested in objects  $X$  of  $\mathcal{C}$  endowed with a three-step filtration

$$0 = X^3 \subset X^2 \subset X^1 \subset X$$

with specified identifications of the successive quotients

$$X/X^1 \simeq Q, \quad X^1/X^2 \simeq R, \quad X^2 \simeq P.$$

Writing

$$F = X^1, \quad E = X/X^2,$$

one obtains two ordinary extensions

$$0 \rightarrow P \rightarrow F \rightarrow R \rightarrow 0, \quad 0 \rightarrow R \rightarrow E \rightarrow Q \rightarrow 0. \quad (9.3.2)$$

Conversely, suppose that the two extensions  $F$  and  $E$  in (9.3.2) are fixed. A *mixed extension* of  $E$  by  $F$ , or in Grothendieck's terminology an *extension panachee*, is such a filtered object  $X$ , together with isomorphisms

$$X^1 \simeq F, \quad X/X^2 \simeq E, \quad (9.3.3)$$

compatible with the filtrations and with the common quotient  $R$ .

For fixed  $P, Q, R, E, F$ , the mixed extensions of  $E$  by  $F$  form a groupoid, denoted

$$\text{EXTPAN}(E, F).$$

Every arrow is invertible, and the automorphism group of any object  $X$  is canonically

$$\text{Hom}(Q, P). \quad (9.3.4)$$

Indeed an endomorphism of the underlying object of  $X$  which is the identity on the graded pieces differs from the identity by the composite

$$X \longrightarrow Q \longrightarrow P \longrightarrow X.$$

The groupoid  $\text{EXTPAN}(E, F)$  is naturally a torsor under the category-group  $\text{EXT}(Q, P)$  of extensions of  $Q$  by  $P$ . The action is the mixed analogue of the Baer sum. If  $G$  is an extension of  $Q$  by  $P$  and  $X$  is a mixed extension of  $E$  by  $F$ , then

$$(G, X) \longmapsto G \wedge X \quad (9.3.5)$$

is obtained by applying the Baer composition to the corresponding extensions of  $Q$  by  $F$ . For each fixed  $X$ , the functor

$$G \longmapsto G \wedge X$$

from  $\text{EXT}(Q, P)$  to  $\text{EXTPAN}(E, F)$  is an equivalence, and one has the associativity relation

$$G \wedge (G' \wedge X) = (G \wedge G') \wedge X. \quad (9.3.7)$$

It follows that

- a)  $\text{EXTPAN}(E, F)$  is a groupoid, and the automorphism group of each object is canonically  $\text{Ext}^0(Q, P) = \text{Hom}(Q, P)$ ;
- b) the set  $\text{Extpan}(E, F)$  of isomorphism classes is either empty or is a torsor under  $\text{Ext}^1(Q, P)$ ;
- c) there is a canonical obstruction class

$$e(E, F) \in \text{Ext}^2(Q, P), \quad (9.3.9)$$

whose vanishing is necessary and sufficient for the existence of a mixed extension of  $E$  by  $F$ .

The class is described as follows. A mixed extension of  $E$  by  $F$  is equivalently an extension of  $E$  by  $P$  whose image in  $\text{Ext}^1(R, P)$  is the class of  $F$ . In the exact sequence obtained from the second sequence in (9.3.2),

$$\text{Ext}^1(E, P) \longrightarrow \text{Ext}^1(R, P) \xrightarrow{\partial} \text{Ext}^2(Q, P),$$

one has

$$e(E, F) = \partial([F]). \quad (9.3.10)$$

The construction is functorial for exact functors. If

$$r : \mathcal{C} \longrightarrow \mathcal{C}' \quad (9.3.11)$$

is exact, and if primes denote images under  $r$ , then there is a commutative diagram of groupoids, up to canonical isomorphism,

$$\begin{array}{ccc} \text{EXT}(Q, P) \times \text{EXTPAN}(E, F) & \longrightarrow & \text{EXTPAN}(E, F) \\ \downarrow & & \downarrow \\ \text{EXT}(Q', P') \times \text{EXTPAN}(E', F') & \longrightarrow & \text{EXTPAN}(E', F'). \end{array} \quad (9.3.12)$$

Assume  $\text{EXTPAN}(E, F) \neq \emptyset$  and choose  $X \in \text{EXTPAN}(E, F)$ . For a mixed extension  $Y'$  of  $E'$  by  $F'$ , one obtains a class

$$e(Y') \in \text{Ext}^1(Q', P') / \mathfrak{S} \text{Ext}^1(Q, P). \quad (9.3.13)$$

It is independent of the choice of  $X$ , and vanishes precisely when  $Y'$  is isomorphic to the image by  $r$  of a mixed extension of  $E$  by  $F$ .

#### 9.4. A finite-level specialization criterion

Let  $S$  be a henselian trait with generic point  $\eta$ , let  $n > 0$ , and put  $\Lambda = \mathbb{Z}/n\mathbb{Z}$ . Let  $\mathcal{C}$  be the category of  $\Lambda$ -modules on  $S$ , for the fpqc topology, and let  $\mathcal{C}_\eta$  be the corresponding category on  $\eta$ . Let  $P, R, Q$  be objects of  $\mathcal{C}$ , with  $E$  an extension of  $Q$  by  $R$  and  $F$  an extension of  $R$  by  $P$ . Assume that  $Q$  is locally free of finite type over  $\Lambda_S$ , and that  $P$  is represented by a finite multiplicative-type group over  $S$ , say

$$P = D(Q^*), \quad (9.4.0)$$

where  $D$  denotes Cartier duality. Suppose a mixed extension  $X_\eta$  of  $E_\eta$  by  $F_\eta$  is given. We shall attach to it a canonical pairing

$$e(X_\eta) : Q_\eta \otimes Q_\eta^* \longrightarrow \Lambda_\eta. \quad (9.4.1)$$

If  $S$  is strictly local, then  $Q$  and  $Q^*$  are constant, and giving (9.4.1) is the same as giving an element of

$$Q^\vee \otimes (Q^*)^\vee. \quad (9.4.2)$$

Apply the construction of 9.3 to the restriction functor  $\mathcal{C} \rightarrow \mathcal{C}_\eta$ . The obstruction group  $\text{Ext}^2(Q, P)$  vanishes, so mixed extensions over  $S$  exist. The required identifications rest on the following computations.

**Lemma 9.4.3.** Under the preceding hypotheses, with  $S$  strictly local, one has canonical isomorphisms

$$\begin{aligned} \text{Ext}^i(Q, P) &\simeq H^i(S, Q^* \otimes P), \\ \text{Ext}^i(Q_\eta, P_\eta) &\simeq H^i(\eta, Q^* \otimes P), \end{aligned}$$

and

$$Q^* \otimes P \simeq D(Q \otimes Q^*).$$

For every finite constant group  $H$  on  $S$ ,

$$H^i(S, D(H)) = H^i(\eta, D(H)) = 0 \quad (i \geq 2),$$

and there is a canonical isomorphism

$$H^1(\eta, D(H))/\Im H^1(S, D(H)) \simeq H^\vee.$$

The proof is obtained from Cartier duality and the Kummer sequence. In the case  $H = \mathbb{Z}/m\mathbb{Z}$ , the cohomology sequence associated with

$$0 \longrightarrow \mu_m \longrightarrow \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m \longrightarrow 0$$

reduces the assertion to the vanishing of the relevant cohomology of  $\mathbb{G}_m$  over a strictly henselian trait and over its generic point. This also describes explicitly the boundary homomorphism giving the last isomorphism.

Combining these computations with 9.3 gives a canonical isomorphism

$$\text{Ext}^1(Q_\eta, P_\eta)/\Im \text{Ext}^1(Q, P) \simeq Q \otimes Q^*. \quad (9.4.5)$$

The class attached in 9.3 to the generic mixed extension  $X_\eta$  therefore becomes the pairing (9.4.1). Its vanishing is exactly the obstruction to extending  $X_\eta$  to a mixed extension over  $S$ . The construction over a merely henselian trait is obtained by descent from the strict henselization.

**Proposition 9.4.7.** With the same data, without assuming that the trait is henselian, the restriction functor

$$\text{EXTPAN}(E, F) \longrightarrow \text{EXTPAN}(E_\eta, F_\eta)$$

is fully faithful. Its essential image consists of those generic mixed extensions for which the pairings (9.4.1), at all closed points after henselization, vanish.

The proof is local for the étale topology on  $S$ . Full faithfulness follows because homomorphisms extend uniquely from the generic point to  $S$ , using normality and finiteness of the Cartier-dual group. Essential surjectivity is reduced to the strictly local case, where it is precisely the obstruction computation above.

## 9.5. Mixed extensions of Barsotti-Tate pro-groups

Fix a prime  $\ell$ . Let

$$P, \quad R, \quad Q \tag{9.5.1}$$

be pro- $\ell$ -Barsotti-Tate groups on  $S$ , and suppose given extensions

$$0 \rightarrow P \rightarrow F \rightarrow R \rightarrow 0, \quad 0 \rightarrow R \rightarrow E \rightarrow Q \rightarrow 0. \tag{9.5.2}$$

For every  $n \geq 0$  this gives finite-level extensions

$$0 \rightarrow P(n) \rightarrow F(n) \rightarrow R(n) \rightarrow 0, \quad 0 \rightarrow R(n) \rightarrow E(n) \rightarrow Q(n) \rightarrow 0. \tag{9.5.3}$$

Assume  $Q$  is of étale type and  $P$  of multiplicative type. If over  $\eta$  a pro- $\ell$ -Barsotti-Tate group  $X_\eta$  is given as a mixed extension of  $E_\eta$  by  $F_\eta$ , then each finite level  $X_\eta(n)$  gives a pairing of the form

$$Q(n) \otimes Q^*(n) \longrightarrow \mathbb{Z}/\ell^{n+1}\mathbb{Z}. \tag{9.5.4}$$

These pairings are compatible in  $n$ , hence define a homomorphism of twisted constant  $\ell$ -adic sheaves

$$e(X_\eta) : Q \otimes Q^* \longrightarrow \mathbb{Z}_\ell. \tag{9.5.5}$$

**Proposition 9.5.6.** The restriction functor from mixed Barsotti-Tate extensions over  $S$  to those over  $\eta$  is fully faithful. Its essential image consists of the generic mixed extensions for which the pairing (9.5.5) vanishes. If  $\text{char } K = 0$ , so that Tate's theorem on homomorphisms of Barsotti-Tate groups applies, then a generic mixed extension belongs to the essential image as soon as its underlying Barsotti-Tate group extends over  $S$ .

The first assertion follows level by level from Proposition 9.4.7. For the last assertion, extend the generic Barsotti-Tate group  $X_\eta$  to  $X$  over  $S$ . Tate's theorem extends uniquely the homomorphism  $X_\eta \rightarrow Q_\eta$ . Its finite-level maps are epimorphisms, so their kernels form a Barsotti-Tate group  $F$ , and the generic isomorphism with the prescribed  $F_\eta$  extends. The quotient by  $P$  is similarly identified with  $E$ , giving a mixed extension over  $S$ .

When  $\ell \neq p$ , all these Barsotti-Tate groups are equivalently twisted constant  $\ell$ -adic sheaves, or representations of  $\text{Gal}(\bar{K}/K)$  on finite free  $\mathbb{Z}_\ell$ -modules. The homomorphism (9.5.5) is then exactly the homomorphism measuring the inertia action, as described in 9.2.



## 9.6. Definition in the semi-stable case

We now apply the preceding construction to the semi-stable abelian variety  $A_K$ . Let

$$P = T_\ell(T), \quad Q = M', \quad R = T_\ell(A^0), \quad (9.6.1)$$

and let  $F$  be the resulting extension of  $R$  by  $P$ . The dual extension attached to  $A'_K$ , after Cartier duality and the orthogonality theorem, gives an extension  $E$  of  $Q$  by  $R$ . As generic mixed extension one takes

$$X_\eta = T_\ell(A_K), \quad (9.6.2)$$

with its canonical three-step filtration induced by the toric and fixed parts. We are therefore in the situation of 9.5, and the pairing (9.5.5) takes precisely the form (9.1.2). For  $\ell \neq p$  it agrees with the inertia construction of 9.2. Under Tate's theorem, it may be described entirely in terms of the Barsotti-Tate pro- $\ell$ -group  $T_\ell(A_K)$  over the generic point.

## 9.7. Descent from a finite semi-stable extension

If  $A_K$  is not assumed semi-stable, choose a finite Galois extension  $K'/K$  such that  $A_{K'}$  has semi-stable reduction over the normalization  $S'$  of  $S$ . Descending the monodromy pairing from  $K'$ , and normalizing by the ramification index

$$e(S'/S) = \text{length}(V'/\mathfrak{m}V'),$$

one obtains a pairing

$$u_\ell : M_K \otimes M'_K \longrightarrow \mathbb{Q}_{\ell,K} \quad (9.7.1)$$

which is independent of the chosen semi-stable extension. When  $A_K$  already has semi-stable reduction, it is the restriction to the generic point of (9.1.2). The integrality theorem below shows that this pairing is in fact independent of  $\ell$  and has values in  $\mathbb{Q}_K$ . In the elliptic curve case the pairing can be interpreted as an ordinary rational number; Neron's results identify it with the valuation of the absolute invariant  $j$ .

# 10. Properties of the monodromy pairing. Integrality and positivity

This section records the formal properties of the monodromy pairings and states the deeper integrality and positivity theorem. The verifications of the formal properties are straightforward from the definitions and are left mostly to the reader.

## 10.1. Transport of structure

The definition of (9.1.2) is compatible with transport of structure in the triple  $(S, A_K, A'_K)$ . Thus every isomorphism of such data carries the corresponding modules  $M, M'$  and the pairing  $u_\ell$  to the corresponding objects for the transported data.

## 10.2. Symmetry

Apply the construction to the dual abelian variety  $A'_K$ . Its dual is canonically identified with  $A_K$ . The resulting pairing

$$u_\ell^{A'} : M'_\ell \otimes M_\ell \longrightarrow \mathbb{Z}_\ell \quad (10.2.1)$$

is obtained from  $u_\ell^A$  by the transposition isomorphism

$$M'_\ell \otimes M_\ell \xrightarrow{\sim} M_\ell \otimes M'_\ell.$$

In other words,

$$u_\ell^{A'}(x', x) = u_\ell^A(x, x'). \quad (10.2.2)$$

### 10.3. Change of trait

Let  $S' \rightarrow S$  be a dominant morphism of traits, and let  $A_{K'}$  be obtained from  $A_K$  by base change. The monodromy pairing changes by multiplication with the ramification index:

$$u_\ell^{S'} = e(S'/S) u_\ell^S. \quad (10.3.4)$$

This is compatible with the description by inertia when  $\ell \neq p$ , and in the general case follows by the finite-level mixed extension construction.

**Theorem 10.4.** Let  $S$  be a henselian trait and let  $A_K$  have semi-stable reduction over  $S$ . Let  $M$  and  $M'$  be the character lattices introduced in 9.1. Then:

a) There exists a unique pairing

$$u : M \otimes M' \longrightarrow \mathbb{Z} \quad (10.4.1)$$

whose  $\ell$ -adic extension is the pairing  $u_\ell$  of (9.1.2), for every prime  $\ell$ . This pairing is separating.

b) If  $\xi : A_K \rightarrow A'_K$  is a polarization, and if

$$(x, y) = u(x, \xi y) \quad (10.4.2)$$

denotes the induced bilinear form on  $M$ , then this form is symmetric, positive, and non-degenerate.

Equivalently, after choosing a geometric point of  $S$ , the induced form on the fibre  $M_0$  is a symmetric positive non-degenerate integral form. The pairing (10.4.1) is again called the monodromy pairing.

### 10.5. Reduction of the proof

The uniqueness in Theorem 10.4 is immediate, and existence may be checked after any dominant change of traits. It is therefore enough to prove the theorem after replacing  $S$  by a complete strictly local trait. If  $A_K$  is isogenous to a direct factor of an abelian variety for which the theorem is known, then the theorem is known for  $A_K$ ; this follows from functoriality of the lattices and of the monodromy pairing, together with Poincaré reducibility.

After a finite extension of  $K$ , every abelian variety is a quotient of the Jacobian of a proper smooth geometrically connected curve. By a theorem of Artin, after another finite extension the curve has a regular proper flat model whose geometric special fibre has only ordinary quadratic singularities, i.e. normal crossings of two branches. Thus Theorem 10.4 is reduced to the case of such a Jacobian. The integral formula and positivity will then be verified geometrically in the Picard-Lefschetz calculation of Section 12.

It remains to check that positivity for one polarization implies positivity for every polarization. If  $\xi_0$  and  $\xi$  are two polarizations, their associated homomorphisms satisfy, in  $\text{End}(A_K) \otimes \mathbb{Q}$ ,

$$\xi = \xi_0 a = a^* \xi_0, \quad a = b^* b$$

for some  $b \in \text{End}(A_K) \otimes \mathbb{Q}$ . In terms of the biextensions attached to the polarizations, this gives

$$\mathcal{B}(\xi) = \mathcal{B}(\xi_0) \circ (b, b). \quad (10.5.5.1)$$

Consequently

$$u_\xi(x, y) = u_{\xi_0}(bx, by). \quad (10.5.5.2)$$

Symmetry and positivity are therefore inherited from  $\xi_0$ , and non-degeneracy is equivalent to the separating property of (10.4.1), which is independent of the polarization.

## 11. Connected components of the Neron model: duality and asymptotic behavior

### 11.0. Component groups

Let

$$\Phi = A/A^0 \quad (11.0.1)$$

be the finite étale group of connected components of the Neron model. For each prime  $\ell$ , let  $\Phi(\ell)$  be its  $\ell$ -primary component. The knowledge of  $\Phi$  is equivalent to the knowledge of all the  $\Phi(\ell)$ . Since formation of the Neron model is compatible with strict henselization, the study of  $\Phi$  may be reduced to the strictly local case; the Galois action on the component group is then recovered by transport of structure.

### 11.1. Strictly local description

Assume  $S$  strictly local. Then

$$\Phi(k) \simeq A(k)/A^0(k) \simeq A(S)/A^0(S) \simeq A(K)/A(K)^0. \quad (11.1.1)$$

Here the first equality uses smoothness over the residue field, the second is Hensel's lemma for the smooth group scheme  $A$  over  $S$ , and the third is the equality  $A(S) = A(K)$  for the Neron model. We put

$$A(K)^0 = A^0(S). \quad (11.1.2)$$

For  $n$  prime to  $p = \text{char } k$ , multiplication by  $n$  is étale and surjective on  $A^0$ , hence  $A(K)^0$  is  $n$ -divisible. Therefore

$$\Phi(k)/n\Phi(k) \simeq A(K)/nA(K). \quad (11.1.3)$$

If  $\ell \neq p$ , let

$$U = T_\ell(A_K(\bar{K})). \quad (11.1.5)$$

Since the fixed part of  $A_K$  is described by the inertia invariants of  $U$ , (11.1.3) yields, for  $n$  a power of  $\ell$ ,

$$\Phi(k)/n\Phi(k) \simeq U_n^I/(U^I)_n. \quad (11.1.6)$$

Passing to the direct limit over  $n$  gives:

**Proposition 11.2.** Assume  $S$  strictly local and  $\ell \neq p$ . There is a canonical isomorphism

$$\Phi(\ell) \simeq (U \otimes D_\ell)^I/(U^I \otimes D_\ell), \quad D_\ell = \mathbb{Q}_\ell/\mathbb{Z}_\ell. \quad (11.2.1)$$

Thus the  $\ell$ -primary component group is determined by the Tate module  $U$  and the inertia action.

### 11.3. The duality pairing for $\ell \neq p$

Let  $U' = T_\ell(A'_K(\bar{K}))$ , and let

$$U \otimes U' \longrightarrow T_\ell(\mathbb{G}_m) \quad (11.3.2)$$

be the perfect Weil pairing, compatible with the inertia action. The structure of inertia gives an exact sequence

$$1 \longrightarrow R \longrightarrow I \longrightarrow T \longrightarrow 1 \quad (11.3.3)$$

with  $R$  pro-prime-to- $\ell$ . Put

$$\Phi = (U \otimes D_\ell)^I / (U^I \otimes D_\ell), \quad \Phi' = (U' \otimes D_\ell)^I / (U'^I \otimes D_\ell). \quad (11.3.4)$$

We define a perfect pairing

$$\Phi \otimes \Phi' \longrightarrow \mathbb{Q}_\ell / \mathbb{Z}_\ell. \quad (11.3.5)$$

Consider the exact sequence

$$0 \rightarrow U \rightarrow U \otimes \mathbb{Q}_\ell \rightarrow U \otimes D_\ell \rightarrow 0 \quad (11.3.6)$$

and the corresponding cohomology exact sequence for  $I$ . The group  $\Phi$  is canonically identified with the kernel of the connecting map into  $H^1(I, U)$ , hence with the torsion subgroup of  $H^1(I, U)$ . The same holds for  $U'$ . The standard duality for the cohomology of  $I$ , using the perfect pairing (11.3.2), gives perfect pairings

$$H^0(I, U \otimes D_\ell) \times H^1(I, U')_{\text{tor}} \longrightarrow D_\ell, \quad (11.3.12)$$

and after passage to the limit,

$$H^0(I, U \otimes D_\ell) \times H^1(I, U' \otimes D_\ell) \longrightarrow D_\ell. \quad (11.3.13)$$

The maps occurring in the two exact sequences are transposes of one another under these pairings. Therefore the cokernel on one side and the kernel on the other are canonically dual, which gives the pairing (11.3.5).

### 11.4–11.6. The semi-stable formula for all primes

The preceding description is only for  $\ell \neq p$ . Suppose now that  $A_K$  has semi-stable reduction. For every prime  $\ell$  we shall define perfect pairings

$$\Phi(\ell) \otimes \Phi'(\ell) \longrightarrow \mathbb{Q}_\ell / \mathbb{Z}_\ell. \quad (11.4.1)$$

This is done by expressing  $\Phi(\ell)$  in terms of the Barsotti-Tate group  $T_\ell(A_K)$  and the monodromy pairing.

**Theorem 11.5.** Assume  $S$  henselian and  $A_K$  semi-stable. Let  $\ell$  be any prime, and let

$$u_\ell : M_\ell \otimes M'_\ell \longrightarrow \mathbb{Z}_\ell$$

be the monodromy pairing. Let

$$\bar{u}_\ell : M_\ell \longrightarrow \text{Hom}(M'_\ell, \mathbb{Z}_\ell)$$

be the corresponding homomorphism, and put  $\Phi_u(\ell) = \text{coker } \bar{u}_\ell$ . Then there is a canonical isomorphism

$$\Phi(\ell) \simeq \Phi_u(\ell). \quad (11.5.1)$$

In particular  $\bar{u}_\ell$  is an isogeny, including for  $\ell = p$ . After Theorem 10.4 one may equivalently state the result by means of the integral monodromy pairing  $u : M \otimes M' \rightarrow \mathbb{Z}$ : if  $\Phi_u$  is the cokernel of  $M \rightarrow \text{Hom}(M', \mathbb{Z})$ , then

$$\Phi \simeq \Phi_u. \quad (11.5.3)$$

To construct the isomorphism, we reduce to the strictly local case. Let  $T$  and  $T'$  be the maximal tori, and identify

$$M = D(T), \quad M' = D(T'). \quad (11.6.1)$$

It is enough to define, for each power  $n$  of  $\ell$ , compatible isomorphisms

$$\Phi(n) \simeq \text{Ker}(M_n \xrightarrow{\bar{u}_n} M_n'^\vee). \quad (11.6.4)$$

Let  $A_n$  be the kernel of multiplication by  $n$  on the Neron model. It is flat and quasi-finite over  $S$ . Its fixed and toric parts give a canonical filtration

$$A_n \supset A_n^f \supset A_n^t \supset 0. \quad (11.6.5)$$

The quotient

$$Y = A_n^f / A_n^t \quad (11.6.6)$$

is flat, separated, and quasi-finite, with generic fibre identified with  $(A_K^0)_n^f / (A_K^0)_n^t$ . The formulas from the preceding section give a canonical isomorphism on the generic fibre

$$M_n \xrightarrow{\sim} Y_\eta. \quad (11.6.7)$$

**Lemma 11.6.8.** The group scheme  $Y$  is etale over  $S$ , and (11.6.7) extends uniquely to a morphism

$$M_n \longrightarrow Y \quad (11.6.8.1)$$

which is an open immersion.

The etaleness follows from the special fibre; normality and separatedness extend the morphism, and an etale monomorphism is an open immersion.

Let

$$\Lambda_n = Y \cap A_n^0. \quad (11.6.9)$$

It is a finite open subgroup of  $Y$ , and the composite

$$\Lambda_n \hookrightarrow Y \hookleftarrow M_n \quad (11.6.11)$$

identifies  $\Lambda_n$  with a subgroup of  $M_n$ . It is enough to prove:

**Lemma 11.6.13.** The image of  $\Lambda_n$  in  $M_n$  is exactly the kernel of  $\bar{u}_n$ .

The kernel is the left annihilator of the finite-level monodromy pairing

$$M_n \otimes M_n' \longrightarrow \mathbb{Z}/n\mathbb{Z},$$

which was defined in terms of the two finite extensions and of the generic mixed extension. By Proposition 9.4.7, a subgroup belongs to this annihilator precisely when the corresponding pulled-back mixed extension extends over  $S$ . The subgroup  $\Lambda_n$  has such an extension by construction, so  $\Lambda_n \subset \text{Ker } \bar{u}_n$ . Conversely, if a subgroup  $K_n \subset \text{Ker } \bar{u}_n$  is given, the associated mixed extension extends over  $S$ . The induced morphism into  $A_n$  factors through  $A_n^0$ , hence through  $\Lambda_n$ ; this gives  $K_n \subset \Lambda_n$ . The lemma follows, and with it Theorem 11.5.

**Remarks 11.7.** The preceding construction shows that the finite mixed extensions (11.6.5), and hence the subgroup  $A_n^0 \subset A_n$ , can be reconstructed from the structural data of 9.6 and from the generic mixed extension. The annihilator condition determines the open finite subgroup, and gluing along the common open part recovers the extension. Thus the system of finite mixed extensions is encoded by the Barsotti-Tate mixed extension and the monodromy pairing.

### 11.8–11.10. Asymptotic behavior under finite extensions

Let  $\bar{K}$  be the separable closure of  $K$ , written as the filtered union of finite extensions  $K'$ . For each  $K'$ , let  $S[K']$  be the normalization of  $S$  in  $K'$ , and let

$$\Phi[K'] \tag{11.8.1}$$

be the component group of the Neron model of  $A_{K'}$ . After a sufficiently large finite extension the reduction is semi-stable, and the transition maps on component groups are injective. Put

$$\Psi = \varinjlim_{K'} \Phi[K']. \tag{11.8.2}$$

For each prime  $\ell$ , Theorem 11.5 gives, for large  $K'$ , an identification

$$\Phi[K'](\ell) \simeq \text{Ker}(M \otimes D_\ell \xrightarrow{u(K')} M' \otimes D_\ell). \tag{11.8.3}$$

The transition maps are compatible with the inclusions into  $M \otimes D_\ell$ ; hence one obtains a canonical monomorphism

$$\Psi \hookrightarrow M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.8.5}$$

**Theorem 11.9.** The homomorphism (11.8.5) is an isomorphism.

It remains only to prove surjectivity. The image is described by the kernels in (11.8.3), and the behavior of the monodromy pairing under ramified extension is given by (10.3.4). Thus it is enough to use the elementary fact that, over a strictly local trait, every integer can be made to divide the ramification degree of a suitable finite separable extension. This makes every element of  $M \otimes \mathbb{Q}/\mathbb{Z}$  appear at some finite stage.

**Corollary 11.10.** Under the hypotheses in which the semi-stable reduction and monodromy constructions apply, the group of connected components of the model after passage through all finite extensions has the form

$$\varinjlim_{K'} \Phi[K'] \simeq M \otimes \mathbb{Q}/\mathbb{Z}. \tag{11.10.1}$$

For  $\ell \neq p$  this is already immediate from the inertia description of Proposition 11.2. The delicate part is the  $p$ -primary component, where Theorem 11.5 is needed. The existence of the isomorphism in Corollary 11.10, with the right-hand side understood as the group associated with the toric character lattice, was conjectured by J.-P. Serre.

## SGA 7-I: Exposé IX, continuation

Néron models, the Picard–Lefschetz formula, and analytic comparison

### 12. Néron model of a Jacobian and the Picard–Lefschetz formula

We first recall the following general results of M. Raynaud. They are used in the form needed below, where the base is a strictly local trait.

**Theorem 12.1 (M. Raynaud).** Let  $S$  be a strictly local trait and let  $f : X \rightarrow S$  be proper and flat, with fibres of dimension one, such that

$$H^1(X, \mathcal{O}_X) = 0 \tag{12.1.1}$$

and such that the local rings of  $X$  are factorial. For each maximal point  $x_i$  of the special fibre, put

$$a_i = \mu_i [k(x_i) : k]_{\text{rad}}, \tag{12.1.2}$$

where  $\mu_i$  is the multiplicity of the component through  $x_i$ , and where the radical degree is a power of the characteristic exponent of the residue field. Let  $d$ , respectively  $d^0$ , be the greatest common divisor of the integers  $a_i$ , respectively of their separable parts. Thus  $d = d^0$  when the residue field is perfect. Put

$$P = \text{Pic}_{X/S} : (\text{Sch})/S^{\text{op}} \longrightarrow (\text{Ab}). \tag{12.1.3}$$

Consider the following conditions:

- 1)  $d = 1$ .
- 2)  $X$  has a zero-cycle of degree 1.
- 3) There exists an invertible module  $L$  on  $X \times P^0$  whose corresponding homomorphism  $P^0 \rightarrow \text{Pic}_{X/S}^0 = P^0$  is the canonical inclusion.
- 4)  $X$  is cohomologically flat over  $S$ ; equivalently, it is cohomologically flat in dimension zero, and the canonical map
$$k \longrightarrow H^0(X_s, \mathcal{O}_{X_s})$$
is an isomorphism, with  $d = 1$ .
- 5)  $P$  is representable, equivalently by a smooth group scheme over  $S$ .
- 6)  $P^0$  is representable, equivalently by a smooth group scheme over  $S$ .
- 7)  $P^0$  is separated over  $S$ , i.e. satisfies the valuative criterion of separatedness; equivalently, every section of  $P^0$  over  $S$  which is zero on the generic point is zero.
- 8)  $d^0 = 1$ .

Then one has the chain of implications displayed in Raynaud’s theorem; in particular, when  $d = d^0$ , for example when  $k$  is perfect, the eight conditions are equivalent.

Assume now that  $d^0 = 1$ , so that  $P$  is represented by a smooth group scheme over  $S$ . Then the schematic closure  $E$  of the unit section of  $P$  is an  $S$ -étale subgroup scheme. If  $I$  denotes the set of reduced irreducible components  $X_i$  of the special fibre, one obtains an isomorphism

$$\mathbb{Z}^I / \mathbb{Z} \xrightarrow{\sim} E \tag{12.1.5}$$

by associating to  $i \in I$  the section of  $P$  defined by the invertible sheaf  $\mathcal{O}_X(X_i)$ . Moreover

$$P^0 \cap E = \text{the zero section of } P. \quad (12.1.6)$$

The quotient

$$Q = P/E \quad (12.1.7)$$

is represented by a smooth separated group scheme over  $S$ , and is Néronian in the sense that it satisfies the Néron mapping property on its special fibre. If, in addition, the geometric generic fibre of  $X$  is integral, there is a canonical exact sequence

$$0 \longrightarrow A \longrightarrow Q \xrightarrow{\deg} \mathbb{Z}_S \longrightarrow 0, \quad (12.1.8)$$

where  $A$  is smooth, separated, and of finite type over  $S$ , and is the Néron model of

$$A_\eta = \text{Pic}_{X_\eta/\eta}^0. \quad (12.1.9)$$

The composite  $P^0 \rightarrow P \rightarrow Q$  induces an isomorphism

$$P^0 \xrightarrow{\sim} A^0. \quad (12.1.10)$$

The case of interest for us is the following. Suppose  $X_\eta$  is smooth, so that, in view of (12.1.1), the geometric generic fibre is smooth and connected, and suppose that the special fibre has a zero-cycle of degree 1. Then all the preceding conditions hold. The group  $A_\eta = \text{Pic}_{X_\eta/\eta}^0$  is an abelian scheme on the generic point, and (12.1.8) gives a very explicit construction of its Néron model. In particular (12.1.10) gives

$$A_s^0 \simeq \text{Pic}_{X_s/s}^0. \quad (12.1.12)$$

It follows that  $A_\eta$  has semi-stable reduction exactly when the maximal torus in  $\text{Pic}_{X_s/s}^0$  has the expected semi-stable form, equivalently in the regular case when the only singularities of the geometric special fibre are ordinary quadratic points. This last verification follows from the usual description of the Picard scheme of a proper separable curve over a field and from EGA IV, 21.8.5; it is also an immediate consequence of the geometric monodromy theorem together with the Galois criterion for semi-stable reduction.

Raynaud's proof is cited from the papers used above. We shall need the theorem only in the special situation just described, and from now on we restrict to the case in which the singular points of the special fibre are ordinary quadratic points.

### 12.3. The toric part of the Picard scheme of a singular curve

Let  $C$  be a proper separable curve over a separably closed field  $k$ , and let  $\tilde{C}$  be its normalization. The Picard scheme  $\text{Pic}_{\tilde{C}/k}^0$  has reductive rank zero, hence the maximal torus of  $\text{Pic}_{C/k}^0$  is contained in

$$N = \text{Ker}(\text{Pic}_{C/k}^0 \longrightarrow \text{Pic}_{\tilde{C}/k}^0).$$

By EGA IV, 21.8.5, the group  $N$  is an extension of a torus  $T$ , described below, by a connected unipotent group. Thus  $T$  is canonically the maximal torus of  $\text{Pic}_{C/k}^0$ .

Let  $J$  be the set of irreducible components of  $\tilde{C}$ . For  $x \in C$ , let  $J(x)$  be the set of branches of  $\tilde{C}$  through  $x$ , i.e. the set of points of  $\tilde{C}$  above  $x$ . Put

$$R(x) = \mathbb{Z}^{J(x)} / \mathbb{Z}, \quad (12.3.1)$$



where  $\mathbb{Z} \rightarrow \mathbb{Z}^{J(x)}$  is the diagonal map, and

$$R = \bigoplus_x R(x), \quad (12.3.2)$$

the sum being taken only over points with at least two branches. Then

$$T = \text{Hom}(R, \mathbb{G}_m). \quad (12.3.3)$$

Equivalently, the character group  $M$  of  $T$  is

$$M = \text{Hom}(T, \mathbb{G}_m) = \text{Ker} \left( \bigoplus_x \mathbb{Z}^{J(x)} \longrightarrow \mathbb{Z}^J \right), \quad (12.3.4)$$

where the map is induced by the natural map sending a branch to the component on which it lies.

We shall use only the case in which at most two branches pass through a point. If  $I$  is the set of points with two branches, then, after choosing an order of the two branches at each such point,

$$R \simeq \mathbb{Z}^I, \quad M = \text{Ker}(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J). \quad (12.3.6)$$

Equivalently one can describe  $M$  by the graph  $\Gamma(C)$  whose vertices are the irreducible components and whose edges are the unordered pairs of branches meeting at a singular point. Then

$$M \simeq H_1(\Gamma(C), \mathbb{Z}), \quad R \simeq H^1(\Gamma(C), \mathbb{Z}). \quad (12.3.7)$$

The canonical homomorphism from the branch description to the graph description is induced by the map from the set of branches over the double points to the set of components.

## 12.4. The pairing attached to ordinary double points

We now place ourselves in the situation fixed at the end of 12.2. By (12.1.12) and by the preceding description applied to  $C = X_s$ , the character group  $M$  of the maximal torus  $T_s$  of the special fibre  $A_s^0$  of the Néron model is

$$M = \text{Ker}(\mathbb{Z}^I \longrightarrow \mathbb{Z}^J), \quad (12.4.1)$$

where  $I$  is the set of singular points of  $X_s$ , and  $J$  the set of irreducible components. The map  $\mathbb{Z}^I \rightarrow \mathbb{Z}^J$  is obtained by assigning, to a double point, the difference of the two components which contain the two branches through that point. Thus, after choosing an order  $(C'_i, C''_i)$  of the two branches through  $x_i$ , one has

$$d(x_i) = C'_i - C''_i. \quad (12.4.2)$$

The operator  $d$ , and the isomorphism (12.4.1), depend on these choices. However, if

$$\varphi_{x_i} : M \longrightarrow \mathbb{Z} \quad (12.4.3)$$

is the composite of  $M \rightarrow \mathbb{Z}^I$  with the projection onto the factor corresponding to  $x_i$ , then  $\varphi_{x_i}$  is determined up to sign, and the sign changes exactly when the order of the two branches is reversed. Consequently the bilinear form

$$\varphi_{x_i} \otimes \varphi_{x_i} \in M^\vee \otimes_{\mathbb{Z}} M^\vee \simeq \text{Hom}(M \otimes M, \mathbb{Z}) \quad (12.4.4)$$

is canonically determined by the singular point  $x_i$ . We may therefore put

$$u = \sum_i \varphi_{x_i} \otimes \varphi_{x_i} : M \otimes_{\mathbb{Z}} M \longrightarrow \mathbb{Z}. \quad (12.4.5)$$

This pairing is well-defined, independent of all choices of signs. It is positive definite, being the restriction to  $M \subset \mathbb{Z}^I$  of the standard quadratic form  $\sum_i X_i^2$  on  $\mathbb{Z}^I$ .

The notation  $M$  for the character group of  $T_s$  differs from the notation used earlier, where  $M'$  denoted the corresponding group for the dual abelian variety. Since here  $A_\eta$  is a Jacobian, it has its canonical principal polarization, giving a canonical isomorphism

$$A_\eta \xrightarrow{\sim} A'_\eta. \quad (12.4.6)$$

We shall use this to identify  $A_\eta$  and  $A'_\eta$ , and hence  $M$  and  $M'$ . For every prime  $\ell$ , the monodromy pairing may therefore be viewed as a pairing

$$u_\ell : M \otimes \mathbb{Z}_\ell \times M \otimes \mathbb{Z}_\ell \longrightarrow \mathbb{Z}_\ell. \quad (12.4.7)$$

**Theorem 12.5 (Picard–Lefschetz formula).** Let  $S$  be a strictly local trait, and let  $X$  be regular, projective and flat over  $S$ , with fibres of dimension one. Assume that the geometric generic fibre is smooth and connected, and that the geometric special fibre has no singularities other than ordinary quadratic points. With the notation introduced in 12.4, for every prime  $\ell$  the monodromy pairing (12.4.7) is the scalar extension to  $\mathbb{Z}_\ell$  of the integral pairing (12.4.5).

This proves the monodromy formula used earlier in the case of a Jacobian equipped with its canonical polarization. By the reduction argument already given in the preceding section, this is enough to imply the general statement for abelian varieties.

For  $\ell \neq p$ , the proof is essentially the cohomological Picard–Lefschetz formula for

$$H^1(X_{\bar{\eta}}, \mathbb{Z}_\ell) \simeq \text{Hom}(T_\ell(A_\eta), \mathbb{Z}_\ell). \quad (12.7.1)$$

This formula is proved later in the seminar in a more general situation; in the present case it is exactly the assertion above, once the definition of the monodromy pairing is unfolded. It remains to treat the case  $\ell = p$ . We reduce it to characteristic zero.

The reduction uses the same principle as the proof of the essential tameness theorem for monodromy on  $T_\ell$ . We may suppose  $S$  complete. Let  $W$  be a Cohen  $p$ -ring with residue field  $k$ , and choose a homomorphism  $W \rightarrow V$  inducing the identity on residue fields. Let  $L$  be the set of singular points of  $X_s$ . Consider the formal moduli space  $\mathfrak{S}$  of  $X_s$  over  $W$ , with universal curve

$$\mathfrak{X} \longrightarrow \mathfrak{S}.$$

As in the local deformation theory recalled earlier,  $\mathfrak{S}$  is isomorphic to the spectrum of a formal power series ring  $W[[T_\lambda]]_{\lambda \in L}$ , and in  $\mathfrak{S}$  there is a normal-crossings divisor

$$D = \bigcup_{\lambda \in L} D_\lambda, \quad (12.8.1)$$

where the component  $D_\lambda$  corresponds to smoothing all singularities except the point indexed by  $\lambda$ . The given family over  $S$  is obtained by pullback from the universal family:

$$\begin{array}{ccc} X & \longrightarrow & \mathfrak{X} \\ \downarrow & & \downarrow \\ S & \xrightarrow{g} & \mathfrak{S}. \end{array} \quad (12.8.2)$$

Let

$$P = \text{Pic}_{\mathfrak{X}/\mathfrak{S}}^0. \quad (12.8.3)$$

Since the special fibre is geometrically reduced of dimension one,  $\mathfrak{X}$  is cohomologically flat in dimension zero, and  $P$  is formally smooth over  $\mathfrak{S}$ . We use the representability of this Picard functor by a smooth group object over  $\mathfrak{S}$ . Alternatively, Artin's representability theorem supplies an algebraic space, which suffices for the argument.

Let

$$U = \mathfrak{S} - \text{Supp } D. \quad (12.8.4)$$

Over  $U$ , the universal curve is smooth and  $P_U = \text{Pic}_{\mathfrak{X}_U/U}^0$  is an abelian scheme. Pulling back along  $g : S \rightarrow U$  gives the abelian variety  $A_\eta$  we are studying. The isomorphism (12.1.10) gives, on  $\mathfrak{S}$ ,

$$P^0 \times_{\mathfrak{S}} S \simeq A^0. \quad (12.8.5)$$

Since the special fibre of  $P$  is  $\ell$ -divisible for every  $\ell$ , so is  $P$  itself. We may therefore consider  $T_\ell(P)$  over  $\mathfrak{S}$ ; its pullback to  $S$  is  $T_\ell(A_\eta)$ . Applying the constructions of the previous sections gives a filtration by the fixed part and the toric part

$$T_\ell(P)^f \subset T_\ell(P)^t \subset T_\ell(P). \quad (12.8.6)$$

Over  $U$  the canonical autoduality of the Jacobian gives a pairing

$$T_\ell(P_U) \times T_\ell(P_U) \rightarrow \mathbb{Z}_\ell(1). \quad (12.8.7)$$

For this pairing, the fixed part and the toric part are mutual orthogonals. Thus the quotient  $E = T_\ell(P_U)^t / T_\ell(P_U)^f$  is canonically dual to  $F = T_\ell(P_U) / T_\ell(P_U)^t$ , and extends canonically over  $\mathfrak{S}$  as

$$E = D(F), \quad F = T_\ell(P)^t / T_\ell(P)^f. \quad (12.8.8)$$

We are therefore in the typical situation of a mixed extension. The group  $T_\ell(P_U)$  is a mixed extension of  $E$  by  $F$ , where  $E$  and  $F$  arise from two dual extensions. Schematically the situation is

$$\begin{array}{ccccccc} 0 & \longrightarrow & F & \longrightarrow & R & \longrightarrow & M \longrightarrow 0 \\ & & & & \uparrow & & \\ & & & & T_\ell(P_U) & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & D(M) & \longrightarrow & R^\vee & \longrightarrow & D(F) \longrightarrow 0. \end{array} \quad (12.8.9)$$

The only difference from the earlier discussion is that the resulting pairing takes values not in a canonically fixed copy of  $\mathbb{Z}_\ell$ , but in

$$\Gamma(U, \mathcal{O}_U^*) / \Gamma(\mathfrak{S}, \mathcal{O}_\mathfrak{S}^*) \otimes \mathbb{Z}_\ell,$$

which is identified with  $\mathbb{Z}^L \otimes \mathbb{Z}_\ell$  by the functions defining the components  $D_\lambda$ . This gives a canonical pairing

$$u : M \otimes M \longrightarrow \mathbb{Z}^L. \quad (12.8.11)$$

The monodromy pairing for the given curve over  $S$  is the composite of this pairing with the homomorphism  $\mathbb{Z}^L \rightarrow \mathbb{Z}$  induced by  $g : S \rightarrow \mathfrak{S}$ . The latter homomorphism is given by an element

$$(d_\lambda)_{\lambda \in L} \in \mathbb{Z}^L, \quad d_\lambda \geq 1, \quad (12.8.12)$$

where  $d_\lambda$  is the multiplicity with which the pullback of  $D_\lambda$  appears on  $S$ . For fixed  $\lambda$ , one has  $d_\lambda = 1$  precisely when  $X$  is regular at the corresponding double point. Since regularity is assumed in Theorem 12.5, all  $d_\lambda$ 's are equal to 1. It remains to prove the following local assertion.

**Lemma 12.8.13.** For  $\lambda \in L$ , the composite of (12.8.11) with the projection  $\mathbb{Z}^L \rightarrow \mathbb{Z}$  onto the factor  $\lambda$  is the form  $\varphi_{x_\lambda} \otimes \varphi_{x_\lambda}$  of (12.4.4).

Complete the strictly local ring at the generic point of  $D_\lambda$  and pull the family back to this trait. The new special fibre has exactly one ordinary quadratic singularity. The preceding constructions commute with this localization: the torus, its character group, the monodromy form, and the specialization morphism all localize compatibly. Thus the assertion reduces to Theorem 12.5 in the case where the base is a trait of characteristic zero and the special fibre has a single ordinary double point. This is the classical Picard–Lefschetz case. The reduction proves the theorem.

**Remarks 12.9–12.10.** The reduction just made also reduces to the classical situation in which the special fibre has one double point. In that case there are one or two irreducible components; these correspond respectively to  $\text{rank } M = 1$  and  $\text{rank } M = 0$ , the latter being the case of good reduction for the Jacobian. Without assuming regularity of  $X$ , the same argument gives a more general formula in which the contribution of each double point is multiplied by the integer  $d_\lambda$  of (12.8.12): the monodromy pairing is obtained from

$$\sum_{\lambda \in L} d_\lambda \varphi_{x_\lambda} \otimes \varphi_{x_\lambda}. \quad (12.10.1)$$

## 13. Links with transcendental theory: the complex analytic case

For general facts on analytic spaces we refer to the standard sources. We review only the analytic dictionary needed to compare the preceding algebraic constructions with Hodge-theoretic ones.

### 13.1. Analytic groups over a base

Let  $S$  be a complex analytic space, and let  $A$  be a commutative analytic group over  $S$ . Assume  $A$  is smooth over  $S$ , and put

$$E = \text{Lie}(A/S), \quad (13.1.1)$$

the locally free  $\mathcal{O}_S$ -module pulled back from the relative tangent sheaf along the zero section. We shall use the exponential homomorphism

$$\exp : E \longrightarrow A, \quad (13.1.2)$$

where  $E$  denotes the analytic vector bundle with sheaf of holomorphic sections  $E$ . It is defined locally near the zero section by the standard theory of formal groups in characteristic zero and then extended by multiplicativity. Its image is the open analytic subgroup  $A^0$ , the union of the neutral connected components of the fibres of  $A$ . If  $R = \text{Ker}(\exp)$ , then one has an exact sequence of relative analytic groups

$$0 \longrightarrow R \longrightarrow E \longrightarrow A^0 \longrightarrow 0. \quad (13.1.3)$$

The group  $R$  is  $S$ -étale and is a closed analytic subspace of  $E$  precisely when  $A^0$  is separated over  $S$ .

The exact sequence (13.1.3) is functorial in  $A$  and commutes with arbitrary base change. In this way one obtains a base-change-compatible equivalence between smooth relative analytic groups with connected fibres and pairs  $(E, R)$ , where  $E$  is a locally free analytic vector bundle on  $S$  and  $R \subset E$  is an  $S$ -étale analytic subgroup.

When  $A$  is abeloid over  $S$ , meaning smooth, proper and separated over  $S$  with connected fibres, then  $A = A^0$ , and  $R$  is locally constant. Its rank in a fibre is twice the complex rank of  $E$ . Conversely, an abeloid analytic group is described by such a local system  $R$  together with a complex analytic vector-bundle structure on the real analytic bundle  $R \otimes_{\mathbb{Z}} \mathbb{R}$ .

### 13.2. Formal completion along a fibre

Fix a point  $s \in S$ . For  $n \geq 0$ , let  $S_n$  be the  $n$ -th infinitesimal neighbourhood of  $s$  in  $S$ , and for an analytic space  $X$  over  $S$ , put  $X_n = X \times_S S_n$ . The system  $\widehat{X} = (X_n)_n$  is the formal completion of  $X$  along the fibre  $X_s$ . It is functorial in  $X$ , and formation of  $\widehat{X}$  commutes with finite projective limits.

**Lemma 13.2.1.** Let  $A$  be as in 13.1, and suppose that:

- a)  $A = A^0$ ;
- b)  $R$  is constant in a neighbourhood of  $s$ ;
- c)  $R_s$  generates the complex vector space  $E_s$ .

If  $A'$  is another smooth analytic group over  $S$ , then the natural map from germs of homomorphisms  $A \rightarrow A'$  near  $s$  to homomorphisms of formal completions

$$\varinjlim_{U \ni s} \text{Hom}_U(A_U, A'_U) \longrightarrow \text{Hom}_{\widehat{S}}(\widehat{A}, \widehat{A'}) \quad (13.2.1.1)$$

is bijective.

*Proof.* Using (13.1.3), we may replace a homomorphism  $A \rightarrow A'$  by a homomorphism of the corresponding vector bundles carrying  $R$  into  $R'$ . Passing to formal completions amounts to passing from an  $\mathcal{O}_s$ -linear homomorphism to its completion. The injectivity is clear. For surjectivity, Nakayama's lemma and the hypothesis that  $R_s$  generates  $E_s$  show that any compatible formal homomorphism of vector bundles sends the local system  $R$  into  $R'$  in a neighbourhood of  $s$ . Thus it comes from a genuine germ of homomorphisms.

Consequently, under the hypotheses of the lemma, the analytic group  $A$  is determined, up to unique isomorphism in a neighbourhood of  $s$ , by its formal completion  $\widehat{A}$ .

### 13.3. The analytic Raynaud extension

Let  $A$  be as in 13.1 with  $A = A^0$ . Write  $R_s$  for the lattice in the fibre. After shrinking  $S$  around  $s$ , the group of germs of sections of  $R$  gives an injective homomorphism from the constant group  $R_s$  to  $R$ . Its image is the largest constant subgroup of  $R$ ; call it the fixed part:

$$R^f \subset R, \quad R_s^f \xrightarrow{\sim} R_s^f. \quad (13.3.2)$$

The quotient

$$A^\natural = E/R^f \quad (13.3.3)$$

fits into an exact sequence

$$0 \longrightarrow R/R^f \longrightarrow A^\natural \longrightarrow A \longrightarrow 0. \quad (13.3.4)$$

The morphism  $A^\natural \rightarrow A$  is ét, hence induces an isomorphism

$$\mathrm{Lie}(A^\natural/S) \xrightarrow{\sim} \mathrm{Lie}(A/S) = E. \quad (13.3.5)$$

For every  $n \geq 0$ , it also induces an isomorphism on the  $n$ -th infinitesimal neighbourhood of the fibre, and therefore

$$\widehat{A^\natural} \xrightarrow{\sim} \widehat{A}. \quad (13.3.6)$$

Thus  $A^\natural$  is the analytic counterpart of the group considered algebraically in the Raynaud construction.

Suppose now that a complex torus

$$T_s \subset A_s \quad (13.3.7)$$

is given. Let  $T$  be the constant torus over  $S$  with fibre  $T_s$ . The induced homomorphism of formal completions comes, by Lemma 13.2.1, from a unique germ of analytic homomorphisms

$$T \longrightarrow A^\natural. \quad (13.3.9)$$

After shrinking  $S$ , this homomorphism is defined on all of  $S$ . Its differential identifies  $\mathrm{Lie}(T/S)$  with a locally direct-factor submodule  $E^t \subset E$ :

$$\mathrm{Lie}(T/S) \simeq E^t \subset E. \quad (13.3.10)$$

Let

$$R^t \subset R_s, \quad R^t \subset R, \quad (13.3.11)$$

be the sublattice defined by the torus  $T_s$  and the corresponding constant subgroup. The immersion condition implies, after shrinking  $S$ , that  $T \rightarrow A^\natural$  is a closed immersion. Put

$$B = A^\natural/T. \quad (13.3.12)$$

Then  $A^\natural$  is an extension

$$0 \longrightarrow T \longrightarrow A^\natural \longrightarrow B \longrightarrow 0, \quad (13.3.13)$$

where  $B$  is smooth and separated over  $S$ , with connected fibres, and

$$\mathrm{Lie}(T) = E^t. \quad (13.3.14)$$

The exponential sequence for  $B$  is

$$0 \longrightarrow R/R^t \longrightarrow E/E^t \longrightarrow B \longrightarrow 0. \quad (13.3.15)$$

Since

$$R^t/R^f = (R/R^f)^t \quad (13.3.16)$$

is constant,  $B$  is abeloid near  $s$  precisely when its fibre  $B_s = A_s/T_s$  is abeloid. Therefore every expression of  $A_s$  as an extension of an abeloid analytic group by a complex torus gives, near  $s$ , a unique extension structure of  $A^\natural$  by a torus whose fibre is the given torus.

### 13.4. Mixed Hodge structures

We reformulate the dictionary (13.1.4). Let  $\mathcal{R}$  be the sheaf of sections of  $R$ , and put

$$\mathcal{E}(R) = \mathcal{R} \otimes_{\mathbb{Z}} \mathcal{O}_S. \quad (13.4.1)$$

When  $R$  is locally constant this is a locally free  $\mathcal{O}_S$ -module. The inclusion  $R \subset E$  gives a canonical homomorphism

$$\mathcal{E}(R) \longrightarrow E. \quad (13.4.2)$$

Let  $F$  be its kernel, so that

$$0 \longrightarrow F \longrightarrow \mathcal{E}(R) \longrightarrow E \quad (13.4.3)$$

is exact on the left. In the most useful case the last map is surjective; then

$$0 \longrightarrow F \longrightarrow \mathcal{E}(R) \longrightarrow E \longrightarrow 0 \quad (13.4.4)$$

is exact. The pair  $(E, R)$  is then equivalent to the pair  $(R, F)$ , where  $R$  is locally constant with finitely generated free  $\mathbb{Z}$ -modules as fibres and  $F \subset \mathcal{E}(R)$  is a locally direct-factor submodule satisfying

$$F \cap \overline{F} = 0 \quad (13.4.5)$$

in the usual fibrewise sense. We regard  $\mathcal{E}(R)$  as endowed with the Hodge filtration

$$\mathrm{Fil}^0 \mathcal{E}(R) = \mathcal{E}(R), \quad \mathrm{Fil}^1 \mathcal{E}(R) = F, \quad \mathrm{Fil}^2 \mathcal{E}(R) = 0. \quad (13.4.6)$$

The exact sequence (13.4.3) is functorial in  $A$  and commutes with base change.

Returning to 13.3, set

$$\mathcal{E}^f = R^f \otimes \mathcal{O}_S, \quad \mathcal{E}^t = R^t \otimes \mathcal{O}_S. \quad (13.4.7)$$

These form the increasing weight filtration

$$\mathcal{E}^f \subset \mathcal{E}^t \subset \mathcal{E} = \mathcal{E}(R), \quad (13.4.8)$$

where  $\mathcal{E}^f$ ,  $\mathcal{E}^t/\mathcal{E}^f$ , and  $\mathcal{E}/\mathcal{E}^t$  are to be thought of as of weights  $-2$ ,  $-1$ , and  $0$ , respectively. The functoriality of (13.4.3), applied to  $T \rightarrow A^\natural \rightarrow A$ , gives

$$F(A^\natural) = F \cap \mathcal{E}^f, \quad F(T) = 0. \quad (13.4.10)$$

Consequently the composite  $\mathcal{E}^t \rightarrow E$  induces an isomorphism

$$\mathcal{E}^t/\mathcal{E}^f \xrightarrow{\sim} E^t = \mathrm{Lie}(T). \quad (13.4.11)$$

Assume moreover that  $R_s$  generates  $E_s$ , for instance in the favourable case where  $A_s$  is an extension of an abeloid analytic group by a torus. Then the corresponding map for  $A^\natural$  is surjective, and one has exact sequences

$$0 \longrightarrow F^\natural \longrightarrow \mathcal{E}^f \longrightarrow E \longrightarrow 0, \quad (13.4.12)$$

with

$$F \cap \mathcal{E}^t = 0. \quad (13.4.13)$$

For the quotient  $B = A^\natural/T$ , one has

$$0 \longrightarrow F(B) \longrightarrow \mathcal{E}(B) \longrightarrow \mathrm{Lie}(B) \longrightarrow 0, \quad (13.4.14)$$

and explicitly

$$\mathcal{E}(B) = \mathcal{E}/\mathcal{E}^t, \quad \mathrm{Lie}(B) = E/E^t. \quad (13.4.15)$$

Using (13.4.11) and (13.4.13), one obtains an isomorphism

$$F^\natural \xrightarrow{\sim} F(B). \quad (13.4.16)$$

**Proposition 13.4.17.** In the favourable situation just described, the Hodge filtration and the weight filtration

$$\mathcal{E} \supset F \supset 0, \quad \mathcal{E} \supset \mathcal{E}^t \supset \mathcal{E}^f \supset 0 \quad (13.4.17.1)$$

on  $\mathcal{E} = R \otimes \mathcal{O}_S$  satisfy:

a) the weight filtration comes by tensoring with  $\mathcal{O}_S$  from a filtration

$$R \supset R^t \supset R^f \supset 0$$

by locally constant analytic groups whose fibres are finitely generated free  $\mathbb{Z}$ -modules;

b) one has

$$F \cap \mathcal{E}^t = 0, \quad F + \mathcal{E}^f = \mathcal{E};$$

thus the filtration induced by the Hodge filtration on  $\mathcal{E}^t$  is concentrated in degree  $-1$ , while that induced on  $\mathcal{E}/\mathcal{E}^f$  is concentrated in degree  $0$ ;

c) the filtration induced on

$$\mathcal{E}^t/\mathcal{E}^f = (R^t/R^f) \otimes \mathcal{O}_S$$

by the Hodge filtration corresponds to the analytic abeloid group  $B$  of (13.3.13).

**Remark 13.4.18.** In Deligne's terminology, after restriction to an open set on which  $R$  is locally constant, the preceding proposition says that  $R$ , together with the Hodge and weight filtrations on  $R \otimes \mathcal{O}_S$ , is an analytic family of mixed Hodge structures. This situation was one of the models for Deligne's conjectures on Néron theory in higher dimensions. However, a feature present in the algebraic theory of Section 7 is not visible in this naive analytic description: when  $A$  is abeloid over the complement of a rare analytic subset, the local system  $R/R^t$  should be constant, and more precisely dual to the corresponding toric system for a suitable dual analytic group. This follows algebraically from the duality theorem used earlier. It is not clear from the elementary analytic methods used here whether the analogous assertion, or even unipotence of the local monodromy action, follows without additional hypotheses. Under algebraicity hypotheses the next section shows that the expected behaviour does hold and relates it to the construction of Section 7.



# SGA 7-I. Exposé IX. Néron Models and Monodromy

## Conclusion

### 13.5. The relatively algebraic situation

We now suppose that the analytic group  $A$  comes from a relative group scheme over  $S$ , smooth and separated, whose fibres are extensions of abelian schemes by tori. We denote this relative scheme again by  $A$ . This abuse of notation is justified by the fact that the functor

$$G \longmapsto G^{\text{an}}$$

from relative group schemes of this type to analytic groups over  $S$  is fully faithful. This is a relative GAGA-type assertion. For the notion of a relative scheme, and for the relation between relative schemes over  $S$  and analytic spaces over  $S$ , the references are Hakim's book and the corresponding earlier exposés on relative analytic geometry.

Let  $O$  be the local ring of  $S$  at  $s$ . Near  $s$ , the relative scheme  $A$  is determined by the ordinary scheme  $A_O$  over  $\text{Spec } O$  obtained by localizing  $A$  at  $s$ . The formal completion of  $A_O$  along its special fibre, which is isomorphic to  $A_s = A_0$ , is the same as the formal completion of the analytic group  $A$  along  $A_s$ , after identifying

$$A_n = A \times_S S_n$$

with algebraic schemes over  $S_n$ . In the favourable case where this formal group scheme, viewed over the completed local ring  $\hat{O}$ , is algebraizable, one obtains a group scheme over  $\text{Spec } \hat{O}$ , extension of an abelian scheme  $B_{\hat{O}}$  by a torus. To simplify the statement, assume that  $B_{\hat{O}}$  is projective over  $\hat{O}$ ; this is the case, for instance, when  $S$  is normal at  $s$ .

**Proposition 13.5.1.** *Under the preceding hypotheses, the analytic group  $A^{\natural}$  of (13.3.3) comes, in a neighbourhood of  $s$ , from a relative group scheme determined up to a unique isomorphism. This group scheme is an extension of a projective relative abelian scheme over  $S$  by a torus. If  $A_{\hat{O}}^{\natural}$  denotes the group scheme over  $\hat{O} = \hat{O}_{S,s}$  that it defines, then  $A_{\hat{O}}^{\natural}$ , with its extension structure, is canonically isomorphic to the Raynaud extension introduced above.*

The last assertion follows at once from the first and from 7.2.1, since  $A_{\hat{O}}^{\natural}$  has, along its special fibre, the formal completion prescribed by (13.3.6). To prove algebraicity of  $A^{\natural}$  near  $s$ , one proceeds as in 7.2.1 and is reduced immediately to proving algebraicity of the abelian quotient  $B$  of  $A^{\natural}$ .

**Lemma 13.5.2.** *Let  $B$  be an analytic abeloid group over the analytic space  $S$ . Let  $s \in S$  be a point such that the formal completion  $\hat{B}$  along the fibre  $B_s$  is algebraizable. Thus the spaces  $B_n = B \times_S S_n$  are algebraizable, and  $\hat{B}$ , viewed as a formal scheme over  $\hat{O}_{S,s}$ , comes from an abelian scheme  $B_{\hat{O}}$  over  $\hat{O}$ . Suppose moreover that  $B_{\hat{O}}$  is projective over  $\hat{O}$ . Then  $B$  is projective over  $S$  after replacing  $S$  by a neighbourhood of  $s$ .*

The hypothesis means that the formal analytic space  $\hat{B}$  has a polarization, that is, a compatible system of polarizations  $p_n$  on the  $B_n$ . The conclusion means that there is a polarization  $p$  of  $B$  over a neighbourhood of  $s$ . Let  $P$  be the analytic space over  $S$  of relative polarizations of  $B$ . It represents the functor which assigns to  $S' \rightarrow S$  the set of polarizations of  $B_{S'}$  over  $S'$ . This is an open subfunctor of the relative Néron-Severi functor  $\text{NS}_{B/S}$ . Standard arguments show that  $\text{NS}_{B/S}$ , hence also  $P$ , is represented by an unramified analytic space over  $S$ . The formal polarization gives a formal section of  $P$  at  $s$ . Since  $P$  is unramified and locally quasi-finite over  $S$ , a formal section is induced by an analytic section over a neighbourhood of  $s$ . This analytic section is the desired polarization of  $B$ .

**Remarks 13.5.3.** (a) If the projectivity hypothesis on  $B_{\hat{O}}$  is omitted, one can still conclude that  $B$  is algebraic near  $s$ . One uses an unpublished theorem, independently due to Artin and Deligne, saying that if  $X$  is proper over  $S$  and if the formal completion of  $X$  along  $X_s$  is algebraizable, then  $X$  itself is algebraizable. This reduces matters to the case where  $S$  is reduced. When  $S$  is normal, an abelian scheme over  $\hat{O}$  is projective, and Lemma 13.5.2 applies. The general reduced case is obtained by non-flat descent from the normalization of  $S$ , using the fact that a morphism for which  $\mathcal{O}_S \rightarrow f_*\mathcal{O}_{S'}$  is injective is an effective descent morphism for relative abelian schemes.

(b) The same method should also prove the analogous result for any analytic space  $B$  proper over  $S$ . The more delicate question is whether an analytic space  $X$  proper over  $S$ , whose formal completion along  $X_s$  is algebraizable, is itself algebraizable near  $s$ . Here algebraizability must be understood in a wider sense than in Hakim's book: near  $s$ , the space should come from an algebraic space over the local base, in Artin's sense.

Still under the hypotheses of Proposition 13.5.1, choose for every  $n > 0$  the standard isomorphism

$$\mu_n \simeq (\mathbb{Z}/n\mathbb{Z})_S, \quad m \longmapsto \exp(2\pi i m/n),$$

and, by passing to inverse limits over powers of a fixed prime  $\ell$ , identify the usual Tate twist with the constant  $\mathbb{Z}_\ell$ -system. Applying the exact sequence of  $\text{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$  to the exact sequence (13.1.3) gives the corresponding canonical projective systems. In the present algebraic situation the finite sheaves that occur are represented by relative finite group schemes, and the analytic symbols used earlier therefore acquire an algebraic meaning.

The resulting identifications are compatible with fixed parts and toric parts. If  $A'$  satisfies the same hypotheses and if  $W$  is a biextension of  $(A, A')$  by  $\mathbb{G}_m$ , defining a duality over the open set where  $A$  and  $A'$  are abelian schemes, then the induced pairings on Tate modules agree with the algebraic pairings. After shrinking around  $s$ , the  $\ell$ -adic pairings glue to an integral pairing

$$R \times R' \longrightarrow \mathbb{Z}. \tag{13.5.11}$$

The algebraic orthogonality theorem can then be applied to  $R$ . The pairing (13.5.11) is zero on the fixed and toric products prescribed by the algebraic theory. If  $S$  is nonsingular of dimension one and if, over  $U = S - \{s\}$ , the restrictions of  $A$  and  $A'$  are dual abelian

schemes, the fixed and toric subobjects are mutual annihilators near  $s$ . Since (13.5.11) is a perfect duality over  $U$ , it follows that  $R/R^t$  is locally constant and that the local monodromy action on  $R$  is unipotent of exponent at most two. One also obtains a perfect duality between  $R/R^f$  and the dual toric lattice. Under this identification the homomorphism of (13.3.4) may be written

$$i : M \longrightarrow A^\natural. \quad (13.5.13)$$

This is the form in which it will reappear below.

## 14. Links with transcendental theory: the rigid-analytic case

### 14.1. The canonical homomorphism $M_K \rightarrow G_K$

We assume the foundations of rigid-analytic spaces. Let  $S$  be a complete trait and let  $A_K$  be an abelian scheme over its fraction field  $K$ . Section 8.1 constructs, by descent from the semi-stable case and from the Raynaud extension, an extension

$$0 \longrightarrow T_K \longrightarrow G_K \longrightarrow B_K \longrightarrow 0 \quad (14.1.1)$$

of an abelian scheme  $B_K$  by a torus  $T_K$ . For the dual abelian scheme  $A'_K$  one has similarly

$$0 \longrightarrow T'_K \longrightarrow G'_K \longrightarrow B'_K \longrightarrow 0, \quad (14.1.2)$$

where  $B'_K$  is dual to  $B_K$ . Let

$$M_K = D(T'_K), \quad M'_K = D(T_K) \quad (14.1.3)$$

be the character lattices. By Exposé VIII, the two extension structures are equivalently described by homomorphisms

$$M_K \xrightarrow{p_K} B'_K, \quad M'_K \xrightarrow{p'_K} B_K. \quad (14.1.4)$$

Let  $W_K$  be the biextension of  $(B_K, B'_K)$  by  $\mathbb{G}_m$  that gives the canonical duality, and let  $E_K$  be its inverse image on  $(M_K, M'_K)$ . A trivialization of  $E_K$  is canonically the same as a homomorphism

$$i : M_K \longrightarrow G_K \quad (14.1.5)$$

lifting  $p_K$ , or, symmetrically, as a homomorphism

$$i' : M'_K \longrightarrow G'_K \quad (14.1.6)$$

lifting  $p'_K$ .

Writing a superscript an for the associated rigid-analytic groups, Raynaud defines a

canonical exact sequence

$$0 \longrightarrow M_K^{\text{an}} \xrightarrow{i^{\text{an}}} G_K^{\text{an}} \xrightarrow{e} A_K^{\text{an}} \longrightarrow 0. \quad (14.1.7)$$

The homomorphism  $i^{\text{an}}$  can also be read as an ordinary homomorphism  $i : M_K \rightarrow G_K$ , lifting  $p_K$ . The sequence is the rigid-analytic analogue of (13.3.4), functorial in  $A_K$  and compatible with change of traits.

By this compatibility one reduces to the semi-stable case. There the groups  $T_K, G_K, B_K, M_K, \dots$  are generic fibres of group schemes  $T, G, B, M, \dots$  over  $S$ , and  $G$  is an extension of  $B$  by  $T$ . Let  $\widehat{G}$  be the formal completion of  $G$  along the special fibre. There is a canonical isomorphism

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0, \quad (14.1.8)$$

and, on rigid generic fibres,

$$\widehat{G}^{\text{an}} \xrightarrow{\sim} \widehat{A}^{0,\text{an}}. \quad (14.1.9)$$

There are also canonical open immersions

$$\widehat{G}^{\text{an}} \hookrightarrow G_K^{\text{an}}, \quad \widehat{A}^{0,\text{an}} \hookrightarrow A_K^{\text{an}}. \quad (14.1.10)$$

The map  $e$  is characterized uniquely by the commutative square

$$\begin{array}{ccc} \widehat{G}^{\text{an}} & \longrightarrow & \widehat{A}^{0,\text{an}} \\ \downarrow & & \downarrow \\ G_K^{\text{an}} & \xrightarrow{e} & A_K^{\text{an}}, \end{array} \quad (14.1.11)$$

where the unnamed arrows are those of (14.1.9) and (14.1.10).

**Remarks 14.1.12.** (a) The uniqueness of  $e$  is immediate from connectedness of  $\widehat{G}^{\text{an}}$ ; existence is the non-trivial part and is supplied by Raynaud's rigid-analytic theory. The definition of  $i^{\text{an}}$  is likewise transcendental, although once  $G$  is constructed by formal geometry the resulting  $i$  is an algebraic datum of the form (14.1.5). At this stage no purely algebraic-geometric description is known. The arithmetic characterization below is obtained by reduction to a situation of finite type over  $\mathbb{Z}$ .

(b) The homomorphism  $i$  is canonically associated with a determined trivialization of  $E_K$ , or equivalently with the dual homomorphism  $i'$ .

## 14.2. Relation with monodromy

Let  $n > 0$ . Applying the exact sequence of  $\text{Ext}^1(\mathbb{Z}/n\mathbb{Z}, -)$  to (14.1.7) on the rigid-analytic site over  $K$ , and using that  $M_K^{\text{an}}$  is torsion-free and  $G_K^{\text{an}}$  is  $n$ -divisible, gives an exact sequence of finite group schemes

$$0 \longrightarrow G_K[n] \longrightarrow A_K[n] \longrightarrow M_K/nM_K \longrightarrow 0. \quad (14.2.1)$$

Passing to inverse limits over powers of a prime  $\ell$  yields

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.2.2)$$

The purely algebraic formulas of 8.1.2 and 8.1.7 give the same exact sequence:

$$0 \longrightarrow T_\ell(G_K) \longrightarrow T_\ell(A_K) \longrightarrow T_\ell(A_K)/T_\ell(A_K)^f \simeq M_K \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.2.3)$$

The compatibility follows from (14.1.11). This compatibility, for a fixed  $\ell$ , already characterizes  $i$  transcendently: the map  $e$  identifies the quotients  $T_\ell(A_K)/T_\ell(A_K)^f$  with the  $\ell$ -adic completions of a single lattice  $\text{Ker } e$ , compatibly with the integral structure coming from the algebraic theory. Thus

$$i : M_K \xrightarrow{\sim} \text{Ker } e. \quad (14.2.3 \text{ bis})$$

The same homomorphism reconstructs the monodromy pairing

$$u : M \times M' \longrightarrow \mathbb{Z}. \quad (14.2.4)$$

In the semi-stable case, assume for simplicity that  $M$  and  $M'$  are constant lattices. Pulling the Poincaré biextension on the abelian quotients back by the two homomorphisms from  $M$  and  $M'$  gives a biextension  $E$  of  $(M, M')$  by  $\mathbb{G}_m$ . For  $(m, m') \in M \times M'$ , the fibre  $E_{m, m'}$  corresponds to an invertible  $V$ -module. Choosing a basis defines, independently of the choice, a valuation map

$$v : E_{m, m'}(K) \longrightarrow \mathbb{Z}.$$

A trivialization  $\varphi$  of  $E$  gives

$$(m, m') \longmapsto v(\varphi(m, m')), \quad (14.2.5)$$

and the monodromy homomorphism is this pairing for the splitting attached to  $i$ .

**Remarks 14.2.6.** Thus the rigid-analytic theory recovers integrality and independence of  $\ell$  of the monodromy pairing. Raynaud's theory also recovers its positivity.

**Remark 14.2.7.** Regard  $M \otimes \mathbb{Q}$  as an extension of  $(M \otimes \mathbb{Q})/M$  by  $M$ . Using  $i : M \rightarrow G_K$  gives

$$0 \longrightarrow G_K \longrightarrow G_K^\natural \longrightarrow (M \otimes \mathbb{Q})/M \longrightarrow 0. \quad (14.2.8)$$

Since  $G_K$  is divisible, this is equivalent to the family of extensions (14.2.2). It is therefore defined by algebraic data and is canonically the group introduced in (2.2.3), after identifying its component group with  $(M \otimes \mathbb{Q})/M$ .

### 14.3. An arithmetic characterization of $i$

For every prime  $\ell$ , the extension (14.2.2) is known algebraically. Hence one knows the image of  $i$  under

$$\mathrm{Hom}(M_K, G_K) \longrightarrow \prod_{\ell} \mathrm{Ext}^1(M_K \otimes \mathbb{Z}_{\ell}, T_{\ell}(G_K)). \quad (14.3.1)$$

When  $M_K$  is constant with value  $M$ , this is the homomorphism

$$\mathrm{Hom}(M, G_K(K)) \longrightarrow \prod_{\ell} \mathrm{Hom}(M, H^1(K, T_{\ell}(G_K))) \quad (14.3.2)$$

obtained from the Kummer boundary maps

$$G_K(K) \longrightarrow \prod_{\ell} H^1(K, T_{\ell}(G_K)). \quad (14.3.3)$$

The kernel consists of those elements of  $G_K(K)$  divisible by every integer  $n > 1$ .

**Lemma 14.3.4.** *Let  $V$  be a discrete valuation ring with residue field  $k$  of characteristic  $p > 0$ , of finite type over the prime field. Let  $G_K$  be a finite type group scheme over  $K$ , with zero unipotent rank if  $\mathrm{char} K = 0$ . Then the subgroup of infinitely divisible elements of  $G_K(K)$  is zero. Equivalently, when  $G_K$  is divisible, (14.3.3) is injective.*

*Proof.* After a finite extension of  $K$  and normalization of  $V$ , reduce to the case where  $G_K$  has a composition series with quotients among finite constant groups, abelian varieties,  $\mathbb{G}_m$ , and additive groups. The finite constant and multiplicative cases are immediate. For an abelian variety, pass to the Néron model  $G$  and reduce to proving that the infinitely divisible subgroup of  $G(S)$  is zero. This is true on the special fibre by the usual dévissage and the Mordell-Weil theorem over a field of finite type. The kernel of  $G(S) \rightarrow G(k)$  is filtered by vector groups over  $k$ , killed by  $p$  on successive quotients; intersection over all infinitesimal levels is zero. The additive case is handled similarly.

**Remark 14.3.4.1.** It is essential to require divisibility by all integers, not only by powers of one prime. For a smooth group over a henselian trait, elements in the kernel of reduction are infinitely  $\ell$ -divisible for every  $\ell \neq p$ . The preceding argument shows that the infinitely  $p$ -divisible subgroup is finite, and the full intersection is zero.

When  $k$  has characteristic  $p > 0$  and is of finite type in the absolute sense, the homomorphism  $i$  is therefore determined by its image under (14.3.1), equivalently by the family of extensions (14.2.2). To obtain a general characterization, let  $S = \mathrm{Spec} V$ , with  $V$  a normal noetherian ring, separated and complete for an adic topology. Let  $A$  be a smooth quasi-projective group scheme over  $S$ , with connected fibres, such that the special fibre is an extension of an abelian scheme by an isotrivial torus and such that  $A_U$  is an abelian scheme over a non-empty open  $U \subset S - S_0$ . Section 7 constructs a group scheme  $G$  over  $S$ , an extension of an abelian scheme  $B$  by a torus  $T$ , characterized by an isomorphism of

formal completions

$$\widehat{G} \xrightarrow{\sim} \widehat{A}^0. \quad (14.3.5.1)$$

For every prime  $\ell$  one obtains on  $U$  an exact sequence

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.3.5.3)$$

If  $A'$  satisfies the same hypotheses and  $W$  is a biextension of  $(A, A')$  by  $\mathbb{G}_m$ , inducing a duality over  $U$ , the resulting pairings on Tate modules give the orthogonality theorem and hence the identification

$$T_\ell(A_U)/T_\ell(A_U)^f \simeq M \otimes \mathbb{Z}_\ell. \quad (14.3.5.5)$$

Thus one obtains the extension

$$0 \longrightarrow T_\ell(G_U) \longrightarrow T_\ell(A_U) \longrightarrow M \otimes \mathbb{Z}_\ell \longrightarrow 0. \quad (14.3.5.6)$$

**Conjecture 14.4.** *In the situation described in 14.3.5, there exists a homomorphism*

$$i : M_U \longrightarrow G_U \quad (14.4.1)$$

*such that, for every prime  $\ell$ , the extension (14.3.5.6) is isomorphic to the extension deduced from  $i$  as in (14.2.2). More precisely, there is one and only one way to assign such an  $i$  to every situation of the type of 14.3.5, compatibly with all base changes  $S' \rightarrow S$  arising from continuous homomorphisms  $V \rightarrow V'$  of noetherian adic rings.*

The existence of  $i$ , compatibly with base change, should follow from Raynaud's rigid-analytic theory in its general adic form. For uniqueness, one reduces to the case where  $S_0$  is of finite type over  $\mathbb{Z}$ . Indeed,  $V$  is a filtered inductive limit of normal subrings of finite type over  $\mathbb{Z}$ , and the situation descends, for large index, to such a subring or to its separated completion for the induced adic topology. Since  $S - S_0$  is Jacobson, two possible homomorphisms are equal if they agree at all closed points of  $U$ . Normalizing the closure of such a point gives a complete trait with finite residue field, and the uniqueness result of the preceding one-dimensional case applies. This proves uniqueness, and then base-change compatibility follows from uniqueness, assuming existence.

**Remark 14.5.** It remains to formulate and prove the expected compatibility between the rigid-analytic homomorphism  $i$  and the analogous complex-analytic homomorphism (13.5.13), or better, to find a common generalization of both constructions.

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